Project Design Phase-II
Technology Stack (Architecture & Stack)

Date	21/10/2026
Team ID	Suhani Porwal
Project Name	**Electric Vehicle Market Performance & Insights**
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the technologies, platforms, databases, BI tools, and integrations that you will use to build your Business Intelligence solution. A well-chosen technology stack helps ensure that the solution is scalable, maintainable, secure, and suitable for data analysis and reporting requirements.

S.No	Component	Description	Technology	Justification / Purpose
1	Programming / Query Language	Language or built-in functionality used to filter, transform, aggregate, and analyze data.	Tableau Calculations / Calculated Fields	Tableau Calculated Fields will be used to create calculated measures, KPIs, ratios, aggregations, and business metrics required for EV market analysis without using a separate programming language.
2	Database / Data Warehouse	Used to store and manage the structured business data required for analysis and reporting.	CSV (Excel/CSV Data Source)	The EV dataset will be maintained in a structured CSV format and connected directly to Tableau. This is suitable for an individual project because the dataset can be easily managed and analyzed within Tableau without requiring a separate database or data warehouse.
3	Data Ingestion	Used to load data from source files into the analytical environment.	Tableau Data Connection	Tableau's built-in data connection will be used to connect the EV dataset directly to Tableau, allowing the data to be imported and made available for analysis and visualization.
4	Data Transformation and Business Logic	Tools and logic used to clean, filter, organize, aggregate data and calculate KPIs and business metrics.	Tableau Desktop – Data Source, Filters, Calculated Fields and Parameters	Tableau will be used to prepare and transform the dataset, apply filters, create calculated fields, group data, and develop business logic required to analyze sales, market share, battery technologies, driving range, customer ratings, and other EV metrics.
5	Data Visualization / Analytics	Used to create interactive dashboards, charts, reports, KPIs, and visual analysis.	Tableau Desktop	Tableau will be the primary BI and visualization tool used to create charts, KPI cards, comparisons, trends, and interactive dashboards for analyzing EV market performance and generating business insights.
6	Front-End / User Interface	Interface through which users interact with dashboards, filters, and visual insights.	Tableau Dashboard	The Tableau dashboard will serve as the user interface, allowing users to interact with filters, charts, KPIs, and different views to explore EV market performance, manufacturers, vehicle categories, battery technologies, and customer ratings.